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Comprehensive Decision Framework Combining Price Prediction and Production-Planning Models for Strategic Operation of a Petrochemical Industry

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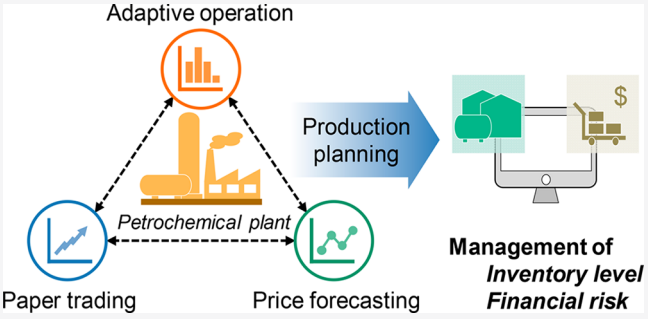
ABSTRACT: This paper presents a new decision-making framework for resource and production planning in the petrochemical industry under price uncertainty conditions. The framework consists of three main decision models: price prediction, paper trading, and production planning. Three different prediction models, system dynamics, multiple linear regression, and artificial neural network, were examined and compared for the precise forecasting of the price of the final products as well as naphtha as a raw material. We then developed a new optimization model using a mixed integer linear programming technique to establish the optimal operational strategies with a given inventory capacity. In the framework, the maximum profitability is ensured by solving a series of decision-making problems on the petrochemical plant, including timing and quantity of naphtha purchase and final product sales, inventory management, financial risk management with paper trading, and operation strategy of the involved processes. To illustrate the capability of the proposed model, the resource and production planning problem of a real petrochemical plant that produces six different major products, ethylene, propylene, butadiene, benzene, toluene, and xylene, was examined. As a result, the optimal production planning using the proposed model enabled the improvement of the total sales by 5.50% and the operating profit by 13.8% in comparison with the BAU case.



1. INTRODUCTION

In the petrochemical industry, intelligent decision making involving resource trade, production planning, resource inventory, and risk management is of paramount importance to ensure the profitability, logistic reliability, and financial safety at the plant and enterprise levels.[1] Furthermore, the prices of raw materials and products have shown astonishing variations in the global petrochemical market, which has one of the world's longest and most complex supply chains. Thus, decision making in a petrochemical plant should cope with both external challenges and internal issues (e.g., energy efficiency and cost reduction) to sustain its competitiveness.

Production planning, as one of the critical strategies to maintain profit margins, covers a wide range of decision problems. These include resource management (including timing and quantities of purchases), product quality management (through efficient operation of the involved processes), and inventory management (according to market conditions). In the petrochemical industry, the importance of appropriate production planning is particularly marked, as its worldwide supply chain is facing a stiff competition, and its markets require a demand-driven model rather than the conventional supply-driven one. A number of studies have attempted to solve the production-planning problems in this industry. In particular, recent reports focus on a better understanding of the nonlinear aspects in petrochemical processes, such as complex catalytic reactions, integrating the supply chain with the petrochemical planning considering various uncertainties (e.g., growing demand for petrochemical products, increasing raw materials costs, and internal or external environment constraints). For instance, a new mathematical optimization model was developed to ensure the maximum business profit in the Norwegian petrochemical industry.[2] The mixed integer linear programming (MILP) technique was used for developing an optimization model that identifies synergic benefits with mergers and acquisitions in the Korean petrochemical industry.[3] Other studies have focused on solving specific problems in the petrochemical industry, including efficient planning and scheduling of the petrochemical

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production with multiple product portfolios, optimal supply chain management, and capacity expansion.[4−6]

To ensure a sustainable operation and management in a petrochemical plant, diverse objectives must have been addressed within the production-planning problems. Such objectives are formulated in the form of an objective function in an optimization model. This not only minimizes the operation costs but also minimizes the raw material requirement for high security, minimizes the harmful environmental impacts, and maximizes the annual profit within a specific regional environment.[7−10] Additionally, the Karimi group developed an agent-based decision support system (DSSs) framework for a supply chain and a petroleum-refinery integrated supply chain modeler and simulator (PRISMS) for a decision-supporting tool for crude oil procurement.[11,12] These systems facilitate integrated decisions in terms of the refinery business. However, their studies only provide insight into responding to changes in domestic policies and unexpected events in the international environment. Göthe-Lundgren, Lundgren, and Persson solved the scheduling problem based on the mixed-integer programming (MIP) model, including utilization of the production process.[13] They formulated an optimization model for production scheduling that can support the decisions of procurement planners at the strategic, tactical, and operational planning levels. However, the developed model is only designed to integrate planning with production scheduling and does not focus on support for procurement or purchase plan decisions. Zhang, Wen, and Xu introduced the mixed integer nonlinear programming (MINLP) technique for optimization of crude oil blending and purchase planning with uncertainty.[14] This is a significant contribution to decision support that considers the short-term process operation and long-term purchase planning simultaneously.

Thus, to maintain the profitability in the aforementioned complex environments with a large uncertainty, the enterprises in the petrochemical industry need to establish intelligent decision making for less-risky, flexible, and adaptive production planning. Whereas the conventional decision-making tools and practices have provided converged solutions for petrochemical plant planning and scheduling problems, a new decision-making framework capable of addressing all the external challenges and internal issues should be proposed. Accordingly, the goal of this study is to develop a novel decision model to support efficient production planning by embedding product price prediction into an optimization-based planning framework to reduce the aforementioned uncertainty, thereby ensuring sustained profit margins.

The proposed framework consists of a financial risk model through precise price prediction for determining timing, quantity, and price of raw material purchases and a plant planning and scheduling risk model to establish the operation strategy of the involved processes to correspond to variable production goals according to the price variation of the final products. The proposed framework is capable of maximizing total revenue by adopting a price prediction model that reduces the business risks for purchase or sales prices in the market. This paper is organized as follows. In Section 2, the optimization problem is described. Section 3 introduces a mathematical model for production planning with the financial risk management technique and for determining the timing of sales of the final products. Section 4 discusses the results of various case studies regarding profit improvement. The final section presents the concluding remarks.

2. PROBLEM DESCRIPTION

A petrochemical plant handles various feedstocks such as representative naphtha, natural gas, and condensate to produce a wide range of primary petrochemicals, including ethylene, propylene, butadiene, benzene, toluene, and xylene. These various petrochemical products are important building blocks in a series of very complex and large-scale chemical networks; hence, the petrochemical industry is highly capital intensive.[15] These primary petrochemicals have an enormous economic significance according to the types of raw materials used. For example, most petrochemical industries in Middle Eastern countries have started to use their own resources to export intermediate or final products with increasing benefits for their economies.[16,17] The petrochemical sector is predicted to be worth approximately USD 600 billion globally and is a very competitive market.[18] Furthermore, the profit margin is very sensitive to the markets of petrochemical products. To maximize the operating profit, a petrochemical company should sell products at their most favorable periods, when high selling prices are available (or predicted). Thereby, the production plan of a petrochemical plant should be established with precise price information on its major products. In this sense, this study proposes an optimal production-planning model, which is capable not only of establishing an optimal operation policy but also of managing the financial risk derived from purchasing raw materials and selling the final products. Accordingly, the decision variables determined by the proposed model are associated with decision-making problems on the petrochemical value chain such as the amount and timing of purchase of raw materials, the inventory levels, and the amount of sales of the final products. Figure 1 shows a typical value chain in the petrochemical industry, from the purchase of raw materials to the sale of final products.

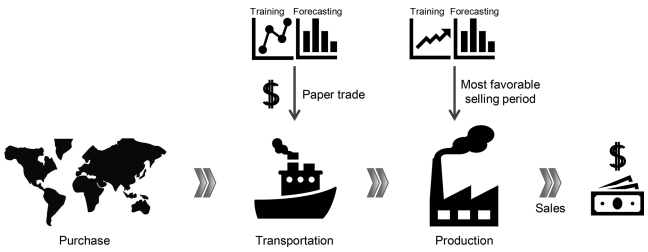

Figure 1. Typical supply chain of the petrochemical industry.

In other aspects, the petrochemical companies need to consider a risk management system to ensure sustained profit margins against the uncertainty of raw material prices. For example, most Korean petrochemical companies import over 90% of their raw materials from the Middle East, and the delivery to destination takes approximately two months. Therefore, they could mitigate a financial risk using one (or multiple) paper trades. As mentioned previously, the petrochemical companies are sensitive to market conditions, which change rapidly. The operating profit of a petrochemical company depends largely on the expenses for purchasing raw materials and the revenue from the sales of the final products. Therefore, it is difficult for most petrochemical companies to establish a precise production plan owing to the changeable

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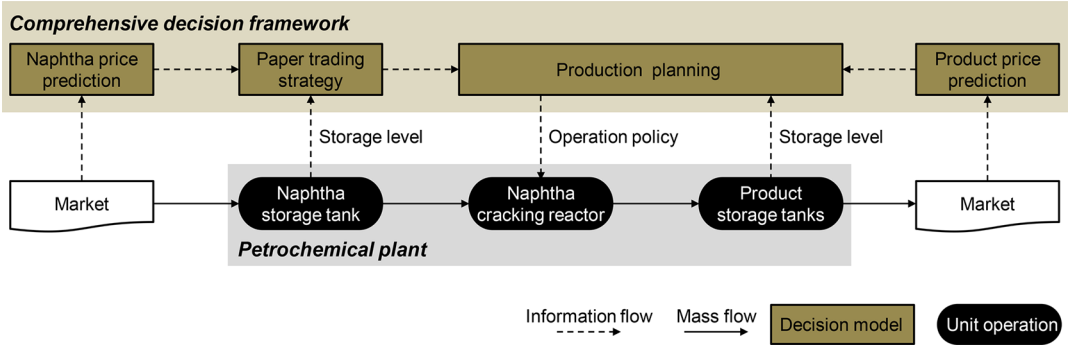

Figure 2. Mass and information flows within major decision-making problems for planning and operation of a petrochemical plant.

prices of raw materials and final products. To handle the financial risk caused by the variation of raw materials and petrochemical prices and to establish sustainable operation strategies, a number of petrochemical companies have adopted a hedge trade system.[19] Hedge indicates that the investment position of the buyer is meant to offset potential gains or losses that might be incurred by forward-looking investment. For instance, most petrochemical companies have been purchasing naphtha (one of their main raw materials) using a hedge trade.[20]

To reflect the aforementioned hedge trade of the petrochemical industry into the proposed framework, this study considers the short-term planning for naphtha purchase, which covers wide-range decision problems such as the number, timing, and quantity of purchases, inventory level management, and operability of the main processes (e.g., production level of a naphtha-cracking unit). This means that a naphtha purchase plan is established by the optimization model to minimize the purchase cost by identifying the optimal timing and quantity of spot trading. The average market prices of naphtha are predicted using a reliable forecasting model proposed in this study. The assumptions in the establishment of the naphtha purchase plan through hedge trading for financial risk management are as follows:

- All of the purchases and sales take place once a month.
- The period of transportation of naphtha from origin to destination is two months.
- Paper trading is realized twice during two months.
- Other financial risk factors are ignored.
- The time interval in the planning is a month.

Therefore, the overall optimization problem consists of determining the following: (1) the timing and quantity of naphtha purchase, (2) the timing and quantity of sales of final products, and (3) the inventory management strategy (tank levels of raw materials and products) to ensure the maximum profit margin from plant operation over a planning period. To solve the aforementioned problems simultaneously, this study integrates different decision models into a single framework as shown in Figure 2. The optimal production planning (e.g., naphtha-cracking yield) is established based on the predicted prices of naphtha and the final products. Further, the optimal timing and quantity of naphtha purchasing and products sales are determined by considering the levels of the storage tanks and the prices for ensuring the maximum profit margin. For instance, the naphtha price prediction using paper trading makes the decision of the maximum naphtha purchase at a lower price to store at a higher level of inventory. Also, the price prediction with the market conditions and storage level of products was considered to set the plant operation. The optimization model determines the decision variables for planning of a petrochemical plant, such as the amount of raw materials required, inventory levels of raw materials and products, amount of sales of the final product, and process unit operability. The model is iterated with initial values (i.e., raw materials and final product inventory levels) until the optimal solutions within upper and lower boundaries are found. Consequently, the operation policy is adapted to operate a naphtha-cracking reactor and subsequent processing for the maximum profit.

3. DEVELOPMENT OF A DECISION MODEL

The proposed comprehensive decision framework consists mainly of two different models: price prediction model for determining precise purchase timing and price, and paper trading for establishing the production planning of the petrochemical plant.

3.1. Price Prediction Model. Price prediction is very important for the determination of the optimal timing of raw material purchasing and product selling. The prediction model of the prices of naphtha and petrochemical products is based on the time series (monthly variations). The prediction model usually includes the following different tasks:

- Step 1: Collect historical data and statistics of price variation behavior.
- Step 2: Select the major factors with effects on the price variation.
- Step 3: Normalize the predicted prices
- Step 4: Analyze quantitatively and qualitatively the predicted price variations compared with the actual price variation.
- Step 5: Identify the correlations between factors and make a casual loop.

Although numerous methods are widely used for price prediction, this study compares three different prediction models for a highly accurate prediction of the prices of naphtha and petrochemical products.[21]

3.1.1. Multiple Linear Regression. The multiple linear regression (MLR) model is one of the most general prediction techniques. Sung et al. predicted the price of naphtha crack considering the naphtha demand in Asia, supply, and margin based on a multiple regression analysis.[22] The MLR model expresses the correlation between two or more input variables and an output variable by fitting a linear equation to the observed data, as shown in eq 1. However, the MLR model mainly deals with quantitative data and has limitations in the

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prediction of phenomena of various fields considering qualitative factors and complexity.

$$P_y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots + \beta_n x_n \tag{1}$$

where P_y is the prediction value as the dependent variable, x_i is the independent variable that affects the prediction value, and β_0 and β_i are parameters.

3.1.2. Artificial Neural Network. The artificial neural network (ANN) is a computational model based on the neural structure of the human brain. The ANN model is applied to various research fields such as time series prediction, signal processing, process control, and heat transfer owing to its advantages for solving complex and nonlinear problems. However, it has serious disadvantages such as overfitting, where prediction data fit well in an observation range but show poor prediction results in other ranges. The ANN model has various architectures, and in this study a feed-forward back-propagation ANN is used. In addition, a sigmoid function is used for the hyperbolic tangent transfer function, which has a range from −1 to 1 in the model. The function $f(x)$ is presented in eq 2.

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{2}$$

3.1.3. System Dynamic Prediction Model. The system dynamics prediction model (SD) is the best approach to understanding the dynamic behavior of complex problems with time using flows, loops, and time delays. A causal loop is a map of a target system with all major factors and their interactions. A causal loop of SD describes the mutual relation between one factor and another factor. In particular, correlations of two or more factors are established with linear or nonlinear mathematical formulations.

3.2. Paper-Trading Model for Production Planning. Paper trading is very important for mitigating the financial risk under fluctuating prices.[23] The paper-trading model is proposed to minimize the average naphtha purchase price of each month and to maximize the operating profit of the petrochemical plant. This model identifies an optimal production plan based on the predicted price of naphtha for three months, the quantity of naphtha used in the cracking reactor, and the inventory level. The model was formulated using a nonlinear programming (NLP) technique.

3.2.1. Objective Function. The objective function is to maximize the total operating profit over a given time horizon as expressed in eq 3. The operating cost includes all the cash flows, where the total sales of each product and profit by paper trade subtracts the operating costs and inventory costs as expressed in eq 4. The objective function is expressed as follows:

$$\text{Maximum profit} = \sum \mathrm{Op}_t \tag{3}$$

$$\mathrm{OP}_t = \mathrm{SP}_t + \mathrm{PP}_t - \mathrm{OC}_t - \mathrm{IC}_t \tag{4}$$

where OP_t, SP_t, PP_t, OC_t, and IC_t denote the operating profit, revenue from total product sales, profitable paper trade using naphtha purchases or sales, operating cost, and inventory cost, whereas the subscript t indicates the planning period. In this model, the sales of products are calculated according to eq 5:

$$\mathrm{SP}_t = \sum_{p \in P} \sum_{t \in T} P_{p,t} \mathrm{ASP}_{p,t} \tag{5}$$

where $P_{p,t}$ represents the prices of product p, and $\mathrm{ASP}_{p,t}$ denotes the sales of product p over a planning period t. PP_t is used to minimize any risk by excess inventory and out of stocks, and it is expressed as follows:

$$\mathrm{PP}_t = \left(\frac{\mathrm{Ap}_{t-1} P_{t-1} + (\mathrm{Pt}_t + \mathrm{Pt}_{t+1}) P_t + \mathrm{Pt}_{t+1} P_{t+1}}{\mathrm{AP}_{t+1} + \mathrm{AP}_{t+2}} \right) \tag{6}$$

where Ap_{t-1}, P_{t-1}, Pt_t, and AP_{t+1} are the determined purchase amounts one month before, price of raw material one month before, purchase or sales quantity by paper trading at planning period t, and total amount of purchases arriving next month.

$$\mathrm{AI}_{r,i,t+1} = \mathrm{AI}_{r,i,t} + \mathrm{AP}_{r,t} + \mathrm{Pt}_t - \mathrm{Pf} \tag{7}$$

where $\mathrm{AI}_{r,i,t}$ is the amount of inventory of raw materials over the planning period t, and Pf is the amount fed into the process.

The operation costs (OC_t) include unloading and harboring costs expressed as eqs 8 and 12. The unloading cost involves the process of unloading the parcel from the vessel to each inventory tank. The harboring cost includes the loading and unloading costs in the port.

$$\mathrm{OC}_t = \mathrm{TUC} + \mathrm{THC} \tag{8}$$

$$U_{r,c,i} = \mathrm{UR}_{r,c,i} \mathrm{UT}_{r,c,i} \tag{9}$$

$$\mathrm{TS}_c = \sum_{r \in R} \sum_{c \in C} \sum_{i \in I} U_{r,c,i} \tag{10}$$

$$\mathrm{TUC} = \mathrm{UT}_{r,c,i} \sum_{r \in R} \sum_{c \in C} \sum_{i \in I} \mathrm{UC}_{r,c,i} \tag{11}$$

where $\mathrm{U}_{r,c,i}$, $\mathrm{UR}_{r,c,i}$, $\mathrm{UT}_{r,c,i}$, $\mathrm{UC}_{r,c,i}$, and TUC are the amount of unloading of raw materials from c to i, transferred unloading rate for raw materials from c to i, required planning during unloading of raw materials from c to i, unloading cost for raw materials from c to i, and total cost over a planning period. TS_c is the total size of cargo on the vessel. In particular, the unloading costs depend on the arrival quantity from purchase spot to plant per month.

$$\mathrm{THC} = \mathrm{UT}_{r,c,i} \sum_{r \in R} \sum_{c \in C} \sum_{i \in I} \mathrm{HC}_{r,c,i} \tag{12}$$

where THC denotes the total harboring cost over a planning period. $\mathrm{HC}_{r,c,i}$ denotes the harboring cost for raw materials from c to i.

The inventory cost (IC_t) is expressed as the sum of the inventory cost for raw materials and final products, as in eq 13:

$$\mathrm{IC}_t = \sum_{r \in R} \sum_{i \in I} \sum_{t \in T} \mathrm{AI}_{r,i,t} \mathrm{PI}_{r,i,t} + \sum_{p \in P} \sum_{i \in I} \sum_{t \in T} \mathrm{AI}_{p,i,t} \mathrm{PI}_{p,i,t}, \forall\, p \in P \tag{13}$$

where $\mathrm{PI}_{r,i,t}$ is the cost of inventory management for raw materials r over a planning period t. $\mathrm{AI}_{p,i,t}$ is the amount of inventory for product p over a planning period t. $\mathrm{PI}_{p,i,t}$ is the cost of inventory management for product p over a planning period t.

Product Yield. The amount of product produced is equal to the amount fed into the cracking process multiplied by the yield of product. The yield equation is expressed as eq 14:

$$\mathrm{AP}_{p,t} = \mathrm{Pf}_t y_p, \forall\, p \in P \tag{14}$$

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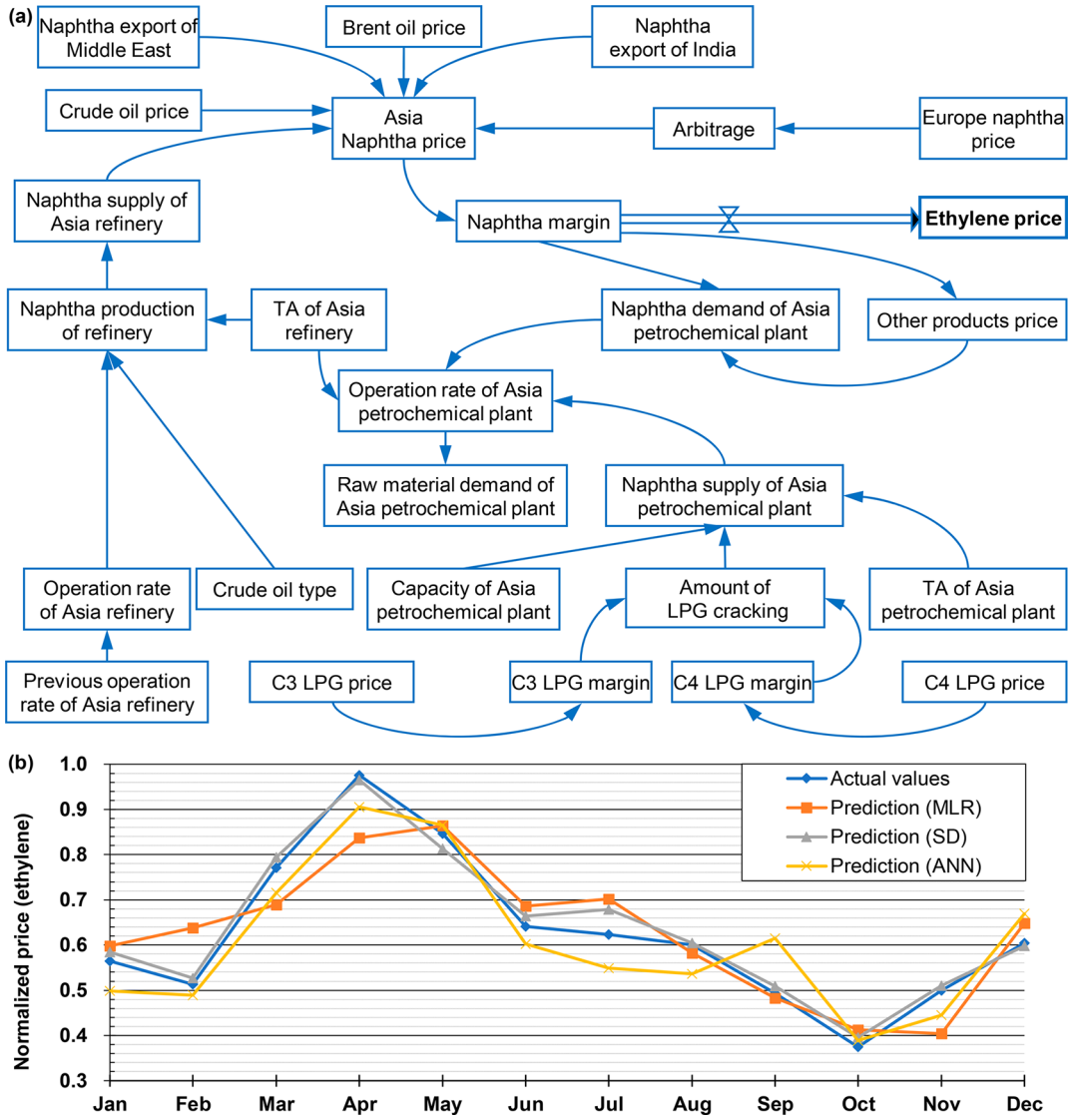

Figure 3. Prediction of ethylene price using the SD model: (a) causal loop for ethylene price prediction and (b) comparison with actual prices and other prediction models.

where $AP_{p,t}$ is the amount of product p produced in the cracking process and y_p is the percent yield of each product p for the specific feed.

$$AI_{p,i,t} = AI_{p,i,(t-1)} + AP_{p,i,t} - ASP_{p,i,t}, \forall\, p \in P \tag{15}$$

where $AI_{p,i,t}$ is the amount of inventory for product p over a planning period t.

3.2.2. Constraints. Demand Constraints. The demands of the final products should be satisfied within a certain range, as shown in eqs 16 and 17:

$$AD_{p,t} \le APS_{p,t} + ALD_{p,t}, \forall\, p \in P \tag{16}$$

$$AD^{L}_{p,t} \le APS_{p,t} \le AD^{U}_{p,t}, \forall\, p \in P \tag{17}$$

where $AD_{p,t}$ is the amount of demand for product p over a planning period t, $APS_{p,t}$ is the amount of product p sold to the global market over a planning period t, $ALD_{p,t}$ is the amount of lost demand for product p over a planning period t, and $AD^{L}_{p,t}$ and $AD^{U}_{p,t}$ are the lower and upper bounds of the demand of product p over a planning period t, respectively.

Availability Constraints. The amount of raw material (naphtha) that should be maintained for stable operation of the petrochemical plant (i.e., naphtha cracking) is as follows:

$$AP^{L}_{t} \le AP_{t} \le AP^{U}_{t} \tag{18}$$

where AP^{L}_{t} and AP^{U}_{t} are the lower and upper bounds of the availability of raw materials r over a planning period t, respectively.

Inventory Constraints.

$$AI^{L}_{r,i} \le AI_{r,i,t} \le AI^{U}_{r,i} \tag{19}$$

$$AI^{L}_{p,i} \le AI_{p,i,t} \le AI^{U}_{p,i}, \forall\, p \in P \tag{20}$$

where $AI^{L}_{r,i}$ and $AI^{U}_{r,i}$ are the lower and upper levels of raw materials r in inventory tank i over a planning period t, and $AI^{L}_{p,i}$ and $AI^{U}_{p,i}$ are the lower and upper levels of product p in inventory tank i over a planning period t.

4. RESULTS AND DISCUSSION

4.1. Price Prediction. To predict the market prices of the six major products from a petrochemical plant (ethylene,

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propylene, butadiene, benzene, toluene, and xylene), this study considered 35 descriptors influencing the petrochemical products. These include historical prices, quantities of demand and supply, price profiles of other reference chemicals (e.g., crude oil), and financial statistics (e.g., margin and cash flow). All the selected factors, which are historical monthly data from May 2009 to December 2010, are used for training the prediction models (MLR, ANN, and SD) to predict future prices from January 2011 to December 2011. Note that this study deals with a short-term prediction for three months in advance involving possible internal and external problems that might occur, including emergency shutdown and unexpected political incidents. However, a majority of the forecasting models published are based on a quantitative approach, which is difficult to apply to real situations. The prices predicted with the SD model were used as raw data for the operation strategy model of the petrochemical plant. Another reason for selecting the SD model in this study is its capability for quantitative and qualitative analysis. Whereas the MLR and ANN models are very powerful to predict the output values (here, future prices), they are limited in providing causality through a quantitative analysis (here, correlation and dynamics between descriptors and to the output). More detailed information about the SD model for the prediction of petrochemical product prices can be found in the Lyu et al. study, which analyzed naphtha demand and supply in a real market using the SD model.[24,25]

In this study, we predicted the prices of major products such as ethylene, propylene, butadiene, benzene, toluene, and xylene using the developed SD model considering time-series data. For instance, Figure 3a and b present the causal loop and the output (ethylene price) and a comparison of the actual and simulation result of the ethylene price using the SD model and other models (MLR, ANN), respectively. Again, this causal loop is helpful to understand the correlation and dynamics between descriptors, such as the correlation between the crude oil price and naphtha margin or between other petrochemical prices and naphtha. Similar to the product price prediction problem, the price of naphtha as a raw material in a petrochemical plant was predicted using the SD model. Notably, the point of naphtha purchase is determined by the fluctuation of the naphtha price.

Table 1 presents the profit (or loss) through paper trading for naphtha purchasing with the predicted price. In seven months, positive cost (i.e., profit) could be achieved. Although there are negative-cost months, the net profit by paper trading in 2011 is approximately 20 million USD. Note that the profit by paper trading is very sensitive to the accuracy of the predicted naphtha price; higher profit can be expected with higher prediction accuracy. In this study, the accuracy of naphtha price prediction is approximately 82.3%, as shown in Figure 3. In particular, the paper trade in the month with relative low prediction accuracy resulted in a negative profit.[26] For instance, the loss by paper trading is the highest in February 2011 owing to the incorrect price prediction and limitation of product inventories (see Figure 3). On the contrary, the profit by paper trading in May, when the actual and predicted prices are almost the same, is the highest.

Table 1. Profitable Cost Using Paper Trade

2011	profitable cost (USD)
January	2 582 991
February	−15 999 139
March	−3 465 293
April	−3 505 513
May	13 684 841
June	6 864 588
July	−5 890 267
August	8 038 669
September	8 514 295
October	7 098 466
November	7 742 833
December	−4 713 173
annual net cost	20 953 299

4.2. Profit Estimation. In this study, the production-planning model in the petrochemical industry is executed to maximize the operating profit using paper trading with the predicted prices of a raw material (naphtha) and the final products. The planning problems are solved using the MILP solver of gPROMS. All the decision variables in the optimization model are determined to ensure the highest operating profit by manipulating a production policy (yield of the naphtha-cracking process). For instance, if the price of ethylene as a main product is expected to increase in the next month, the decision maker operates the plant to maintain the maximum level of the ethylene storage tank by increasing the ethylene yield and/or decreasing the sales this month. Herein, it was assumed that the capacity of the petrochemical plant is 3.6 million ton of naphtha per year. Accordingly, the capacity of the naphtha-cracking process is 0.3 million ton per month, and the storage size of the inventory of the raw materials and products ranges from a minimum of 80 000 to a maximum of 0.3 million tons per tank. The plant specification assumed in this study (e.g., inventory size and naphtha-cracking yield) is summarized in Table 2.

Table 2. Basic Information of the Naphtha-Cracking Plant

	inventory capacity (ton)	product yield
raw material	190 000	
products		
ethylene	20 000	0.33
propylene	15 000	0.16
butadiene	5 000	0.10
BTX[a]	7 500	0.21
others	15 000	0.20

[a]BTX includes benzene, toluene, and xylene.

To illustrate the capability of the comprehensive decision framework, this study examined seven different cases according to whether each decision model (i.e., price prediction, paper trading, and operation strategy) is included or not, as presented in Table 3. Case 1, as the base case, represents the production planning without any proposed decision model, whereas Case 7 indicates the operational profit when the decision framework is fully applied. Figures 4 and 5 present the monthly sales and operating profit, respectively, corresponding to the examined cases. Note that Figures 4h and 5h compare the total annual business sales and operating profit, respectively, of all the examined cases.

In Case 1, where the financial risk model and price prediction were not applied with the fixed product strategy (i.e., no yield change according to product price), the maximum of sales is achieved with the conventional practices. Figures 4a and 5a present the monthly sales and operating

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Table 3. Cases of the Optimization Problem in This Study

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
operation strategy	fixed	fixed	fixed	fixed	adaptive	adaptive	adaptive
paper trading	O	O	O	X	X	O	O
price forecasting	X	X	O	O	X	X	O

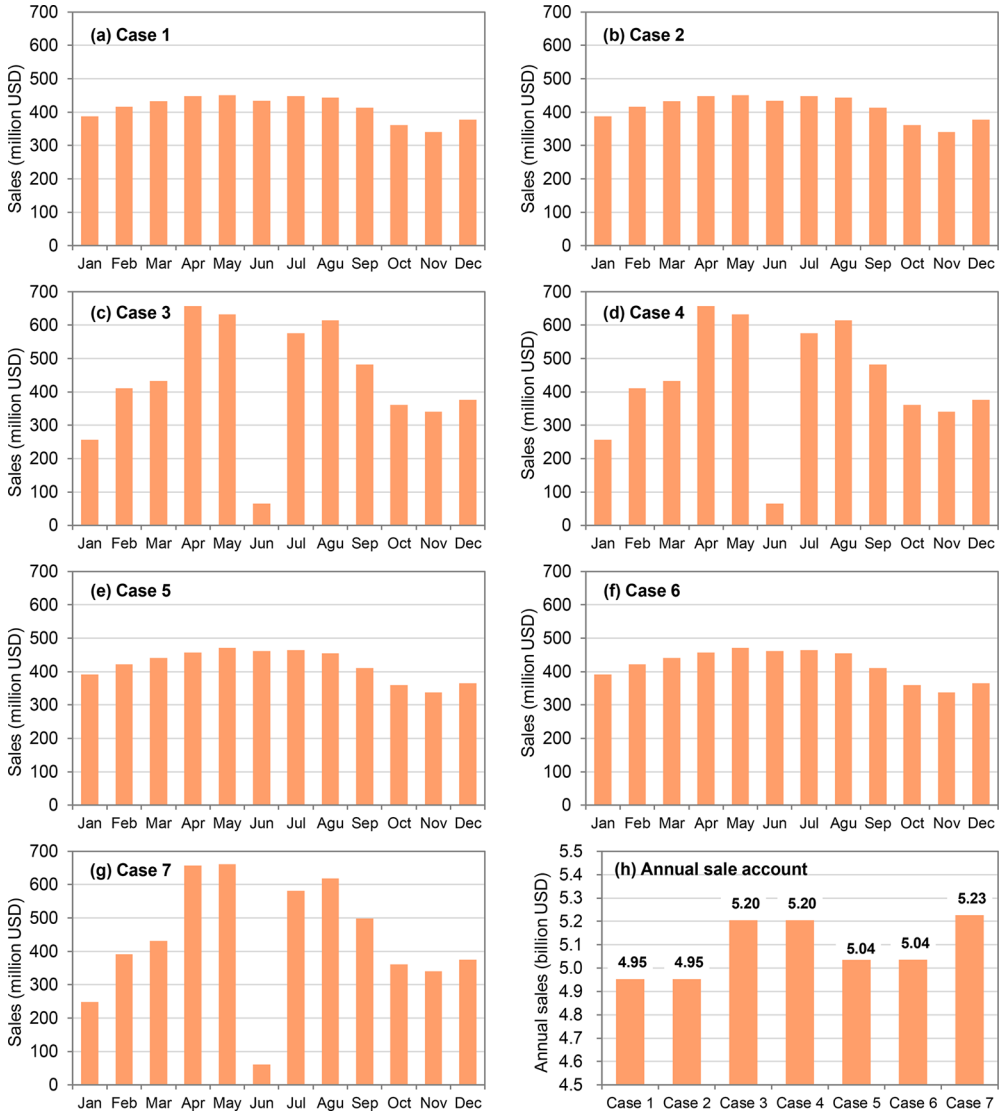

Figure 4. Monthly sales of the case studies.

profit of Case 1, respectively. The amounts of sales and purchases are constant at 0.3 million ton per month. The total sales price is 4.95 billion USD (in Figure 4h), and the operating profit is 297.21 million USD (in Figure 5h), which are obtained from the sales of products (ethylene: 1.118 million tons, propylene: 0.576 million tons, butadiene: 0.360 million tons, BTX: 0.75 million tons, and others: 0.720 million tons).

Figures 4b and 5b show the optimization results of Case 2. Although the total sales account of Case 2 is identical to that of Case 1, a higher operating profit, of 318 million USD, is expected compared to that of Case 1, as shown in Figure 5h. In this case, the additional profit through paper trading is calculated to be approximately 20 million USD. It means that the total operating profit can be improved by 7.04% by applying paper trading for naphtha purchasing.

In addition to the effect of paper trading, Case 3 illustrates the benefits of use of the product price prediction model in production planning. Figures 4c and 5c present the monthly sales and operating profit of Case 3, respectively. The total sales account of Case 3 is 5.20 billion USD, with an increase of approximately 250 million USD over Case 1. Furthermore, the total operating profit is calculated to be 333.21 million USD, an increase of 12.11% compared to Case 1 (Figure 5h), which is higher than the increase attained by paper trading (i.e., 7.04% in Case 2). Thus, it is obvious that production planning with well-predicted prices of the final product can improve the profit. Such benefit can be observed in Case 4, which was included to investigate the net impact of price prediction without paper trading. The results of Case 4, such as monthly sales account and operating profit, are presented in Figures 4d and 5d, respectively. The operating profit is calculated to be

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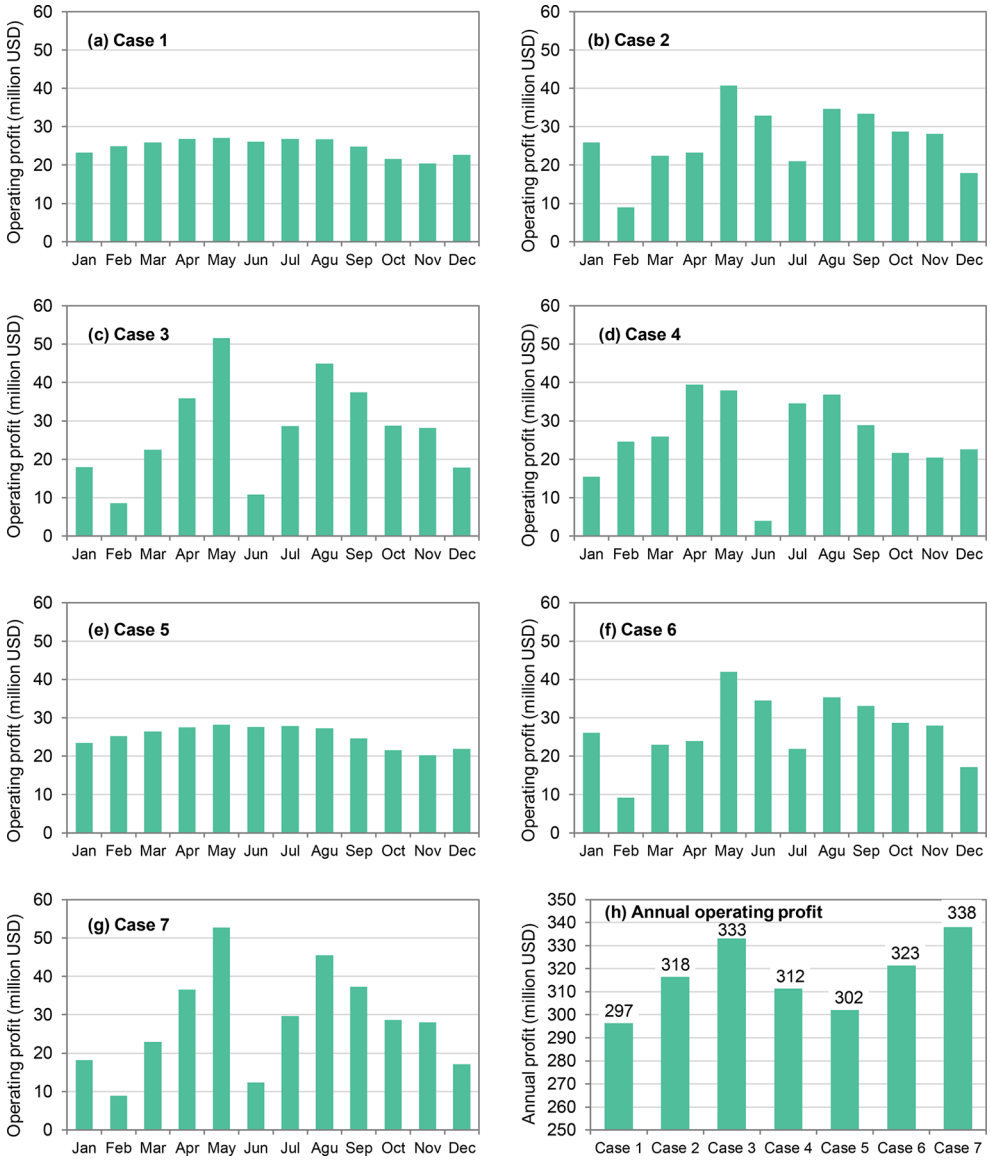

Figure 5. Monthly profit from sales of the products of the case studies.

312.26 million USD. The operating profit improved by 5.06% in comparison with Case 1, as shown in Figure 5h. It is interesting that Case 4 shows the smallest operating profit in June. This is primarily due to the dramatic reduction in product sales over the previous months. The product sales are determined by product price prediction including the levels of each product inventory. For instance, if the price of propylene and other products will increase next month, the decision maker should determine the maximum level of storage for these products. Accordingly, Case 4 implies that inventory management (i.e., storage tank level) is the key strategy to maximize the operating profit under variations of the product prices when there is no corresponding production management strategy (e.g., change of the product portfolio in the naphtha-cracking process).

Adaptive change of the operation strategy (i.e., cracking yield) according to the change in market conditions is one of the very critical options to maximize the operating profit. For instance, if the predicted price of ethylene is expected to increase, the naphtha-cracking process should be operated to maximize the production rate of ethylene. Figures 4e and 5e show the optimal results of the monthly sales (5.04 billion USD) and operating profit (302.13 million USD), respectively, of Case 5, which considers the change in yield of the naphtha-cracking process without a price prediction. Meanwhile, Figures 4f and 5f present the results of Case 6 involving the adaptive operation strategy and financial risk management through paper trading. Case 6 shows an identical sales account to that of Case 5, whereas the total operating profit is expected to be 323.08 million USD, an improvement of 8.70% and 6.94% compared to Cases 1 and 5, respectively.

Finally, Figures 4g and 5g show the results of the case in which the comprehensive decision framework proposed in this study was applied. The achieved total business sales of 5.23 billion USD are the highest compared to all the other examined cases. Furthermore, the total operating profit (338 million USD) is also the highest compared to other cases; in particular, it is 13.8% higher than the base case (Case 1). Thus, it is obvious that the proposed framework, which consists of financial management through paper trading, an adaptive operating strategy, and price prediction models, is effective to improve the profitability of the petrochemical plant.

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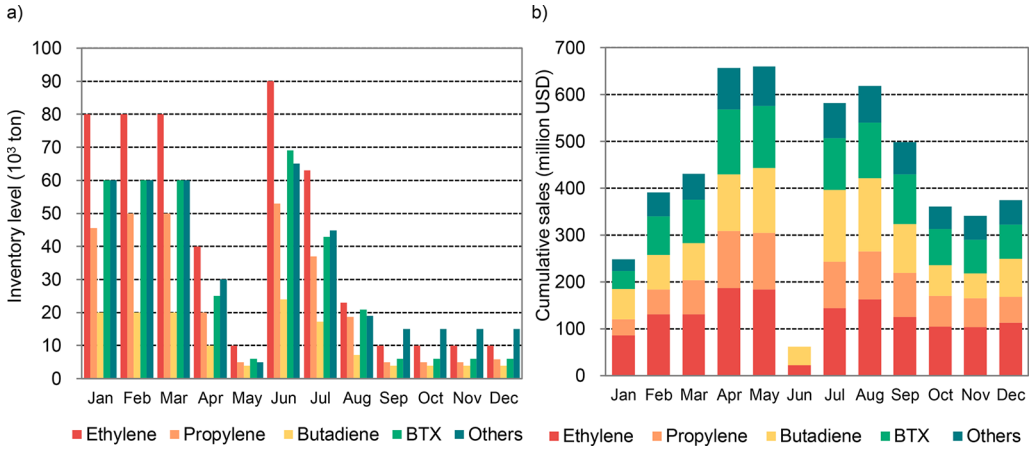

Figure 6. Inventory level (a) and the cumulative sales of the final products in Case 7 (b).

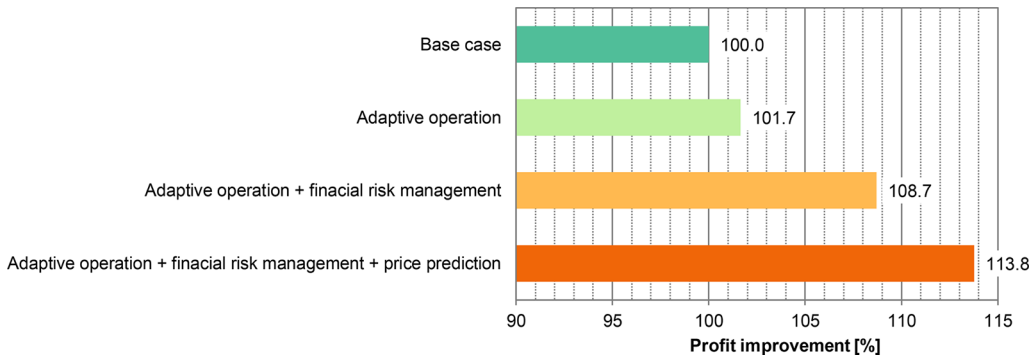

Figure 7. Profit improvement by applying different decision models.

Figure 6 shows detailed information on the inventory level and the cumulative sales of all the major products in Case 7, which results in the highest profitability by fully applying the proposed decision framework. According to the predicted prices of the final product and naphtha as a raw material, the petrochemical plant yields different product portfolios and sells them in the best timing. For instance, sales of ethylene and propylene in April and May are higher than sales in other months, given that the sale prices in the next two months (June and July) are expected to be low. At the same time, the fact that the inventory levels are dramatically decreased corresponds to the high sales in April and May. Meanwhile, the naphtha and final products are expected to store as if the unfavorable sale prices are predicted, given the lower sales in June.

Figure 7 compares the relative improvement in profit obtained by each decision model (i.e., price prediction, paper trading, and operation strategy) compared to the base case (Case 1). Whereas the application of the only adaptive operation model does not generate a huge improvement against the base case (approximately 1.7%), its combination with the paper-trading model creates a huge synergy by efficiently managing the financial risk (8.7% increase over the base case). The proposed comprehensive decision model, which includes adaptive operation, paper-trading, and price prediction models, can achieve an improvement of the probability of 13.8% against the base case.

5. CONCLUSION

This study aimed to develop a new comprehensive decision framework using an optimization technique for the strategic planning and operation of a petrochemical plant. In achieving the goal, this study integrated three different decision models (the adaptive operation, paper-trading, and price prediction models) to maximize the operating profit of the petrochemical plant under variable market conditions. Specifically, the price prediction model using the SD technique forecasts the prices of the major products and of naphtha as the raw material in the petrochemical plant, and the paper-trading model supports the financial management by determining the best timing and quantities for naphtha purchase. The last embedded model, the adaptive operation model, identifies the optimal production strategy of the naphtha-cracking process by considering the storage levels and market prices. As a result, the seven different case studies revealed that the proposed decision model can generate a huge improvement in profitability (13.8%) against the base case where no model was applied.

Although this study provides a useful and efficient decision model for petrochemical plant planning, the proposed framework still has limits as a complete business intelligence, which enables real petrochemical industries to make strategic decisions, improving operational efficiency and business profitability. Accordingly, future research needs to be undertaken to address the following issues: diversification of the petrochemical value chain by including different raw material options and product markets; integration of system-level solutions (e.g., optimal yield of the naphtha-cracking process)

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with process operation policy (e.g., optimal temperature and feed rate of the naphtha-cracking reactor).

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Notes

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NOMENCLATURE

A. Indices

c cargo of vessel
i inventory tank
p petrochemical products
r raw material
t planning time interval

B. Production Planning

$AD_{p,t}$ amount of demand for product p over a planning period t
$AD^U_{p,t}$, $AD^L_{p,t}$ upper and lower bounds on the demand of product p over a planning period t
$AI_{p,i,t}$ amount of inventory for product p over a planning period t
$AI_{r,i,t}$ amount of inventory for raw materials over a planning period t
$AI_{r,i,t-1}$ amount of inventory for raw materials over a planning period $t - 1$
$AI_{r,i,t+1}$ amount of inventory for raw materials over a planning period $t + 1$
$AI^U_{r,i}$, $AI^L_{r,i}$ upper and lower levels r in inventory tank i over a planning period t
$AI^U_{p,i}$, $AI^L_{p,i}$ upper and lower levels of product p in inventory tank i over a planning period t
$ALD_{p,t}$ amount of lost demand for product p over a planning period t
AP_{t-1} purchase amount at a planning period $t - 1$
AP_{t+1} total amounts of purchase at a planning period $t + 1$
$AP_{p,t}$ amount of product p produced from cracking process at a planning period t
$AP_{p,i,t}$ amount of product p produced from cracking process to inventory tank i at a planning period t
$AP_{r,t}$ amount of product p produced from cracking process at a planning period t
AP^U_t, AP^L_t upper and lower bounds on the availability of raw materials r over a planning period t
$APS_{p,t}$ amount of product p sold to global market over a planning period t
$ASP_{p,t}$ sales of product p over a planning period t
$ASP_{p,i,t}$ sales of product p in inventory tank i over a planning period t
$HC_{r,c,i}$ harboring cost for raw materials from c to i
IC_t inventory cost over a planning period t
OC_t operation cost over a planning period t
OP_t operation profit over a planning period t
P_{t-1} price of raw material at a planning period $t - 1$
$P_{p,t}$ price of product p over a planning period t
Pf amount of feed into the cracking process
Pf_t amount of feed into the cracking process per planning period t
$PI_{p,i,t}$ cost of inventory management for product p over a planning period t
$PI_{r,i,t}$ cost of inventory management for raw materials r over a planning period t
PP_t purchase cost over a planning period t
Pt_t purchase or sell quantity by paper trade at a planning period t
Pt_{t+1} purchase or sell quantity by paper trade at a planning period $t + 1$
P_y prediction value as the dependent variable in the multiple linear regression model
SP_t total sales account over a planning period t
THC total harboring cost over a planning period t
TS_c total size of cargo of the vessel
TUC total unloading cost over a planning period t
$U_{r,c,i}$ amount of unloading for raw materials from c to i
$UC_{r,c,i}$ unloading cost for raw materials from c to i
$UR_{r,c,i}$ transferred unloading rate for raw materials from c to i
$UT_{r,c,i}$ required planning during unloading of raw materials from c to i
x_i independent variable that affects the prediction value in the multiple linear regression model
y_p percent yield of each product p for the specific feed
β_0, β_i parameters in the multiple linear regression model

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